Problem 1. ARIMA model and Kalman Filter (40%)

For this problem, use the dataset **Dataset-Stock.txt**. The dataset consists the realized daily volatility series of Alcoa stock returns from January $2, 2003$ to May $7, 2004$. The volatility series is constructed using 20-minute intra-daily log returns.

(a) (10%): Estimate the local trend model in Equations (11.1) and (11.2) in the slide Week 11. Write down the exact equation of your estimated model.

(b) (10%): Obtain time plots for the filtered variables with pointwise 95 % confidence interval. What is the mathematical expression of these “filtered variables”?

(c) (10%): Obtain time plots for the smoothed variables with pointwise 95 % confidence interval. What is the mathematical expression of these “smoothed variables”?

Problem 2. Gibbs Sampling and Markov Switching Model (50%)

For this problem, use the dataset **Dataset-GE.txt**. The dataset consists GE stock monthly log return from 1926 to 1999. We would like to fit a Markov switching model as stated in Week 12, page 63, equation (12.55) and (12.56).

(a) (10%): Do a Gibb sampling to obtain the posterior distribution as in Figure 12.16, page 69-70. You can choose your own prior parameters. Pay special notice how you sample α_{ij}, as stated in page 66.

(b) (10%): Based on the algorithm in (a), present the posterior distribution of state 2 as in the bottom figure on page 68.

(c) (10%): Instead of the Bayesian way in (a) – (b), we can also do it through a Baum-Welch algorithm. Do it a present the esimation for the parameters.

(d) (10%): Based on the algorithm in (b), present the inferred states for each time point.